Post-COVID Math Achievement Gaps

An Examination of Race, Gender, and Socioeconomic Disparities on Education post-COVID

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Why this data matters?

Millions of students are behind academically in both math and reading and recovery attempts are not reaching the students who need it most.

Gaps Compound Over Time	COVID Exposed A Systemic Issue	Resources Are Not Enough
These gaps compound over time leading to poor academic performance in high school, it affects access to high earning degrees and lifetime earnings.	COVID only exacerbated issues that were already present. Students who were most affected by achievement gaps are more affected post-COVID.	Resources alone do not solve the problem without honing in on an equity strategy that is appropriately distributed to students who are the most affected.

Dataset

Dataset: Stanford Education Data Archive (SEDA)
The longitudinal and covariate datasets were combined to capture socioeconomic status.
Grade levels included: 3-8
Subject Focus: Math
Observations: Over 1.7 million observations and 88 variables from the years 2009 to 2024
Pre-COVID: 2017-2019 (n=185,558 observations)
Post-COVID: 2022-2024 (n=162,591 observations)
Standardized: The national average is standardized by SEDA and is based on an average from 2009-2024.

Research Question 1

How does math achievement post-COVID differ across race, gender, and grade level?

Research Question 2

To what extent are post-COVID math achievement gaps by race and gender affected by socioeconomic status, after controlling for grade level and year?

(Reardon et al., 2026)

Data Preparation

Filtering:

- The focus was mathematics for this study so all other subjects were filtered out. Additionally, the focus was an analysis of post-COVID achievement so to compare pre- and post-COVID, 2 groups were created. The pre-COVID group were observations from 2017-2019 and the post-COVID group were observations from 2022-2024.

Achievement Gap Construction

- There were 3 achievement gap variables created: Black - White, Hispanic - White, and Female - Male. These were constructed by subtracting the mean scores of the reference group (White students) from the mean scores of the focal group (Black, Hispanic, and Female).

SES Measure:

- A feature with the lowest missingness (4.7%) was used. It is an index that captured income, parental education level, unemployment, poverty, and the incidence of single parenthood.

Score:

- All scores are expressed in standard deviations. A standard deviation of zero (0) is the national average. 1 standard deviation represents 3 years of learning. Tables show both standard deviations and the number of years to recover. Years to recover means the number of years it would take the student group to reach a standard deviation of (0) or average.

(Reardon et al., 2026)

Methodology

2 OLS regression models were used with district clustered standard errors

Model 1: Post-COVID regression

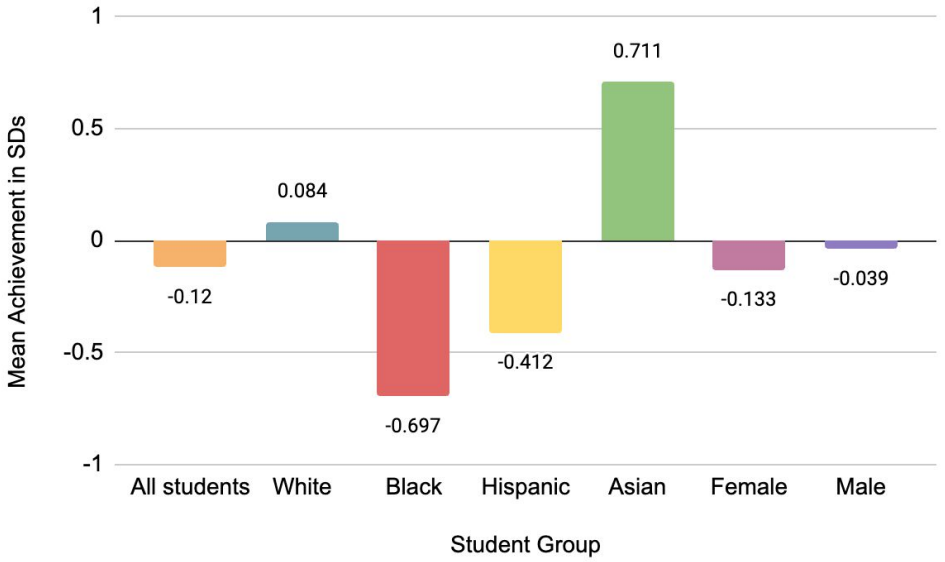
- Post-COVID period focus
- Measures the size of achievement gaps by race, gender, and grade level
- Regression is ran separately for Black-White, Hispanic-White, and Female-Male
- Accounts for grade level and year
- Tells us whether achievement improves or decreases in wealthy and poor districts

Model 2: Pre to Post Comparison

- Predictor variable: ***SES*** was constructed by SEDA and combines 5 indicators to create an index (household income, parental education level, poverty rate, unemployment rate, and single parent household rates).
- Accounts for grade level and year
- ***Compares district averages*** from pre- to post-COVID
- ***Measures change in gap*** from pre- to post-COVID
- Tells us whether wealthier districts recovered better or worse post-COVID

Findings are at the district level and describe patterns across the United States.

Post-COVID Math Achievement by Student Group

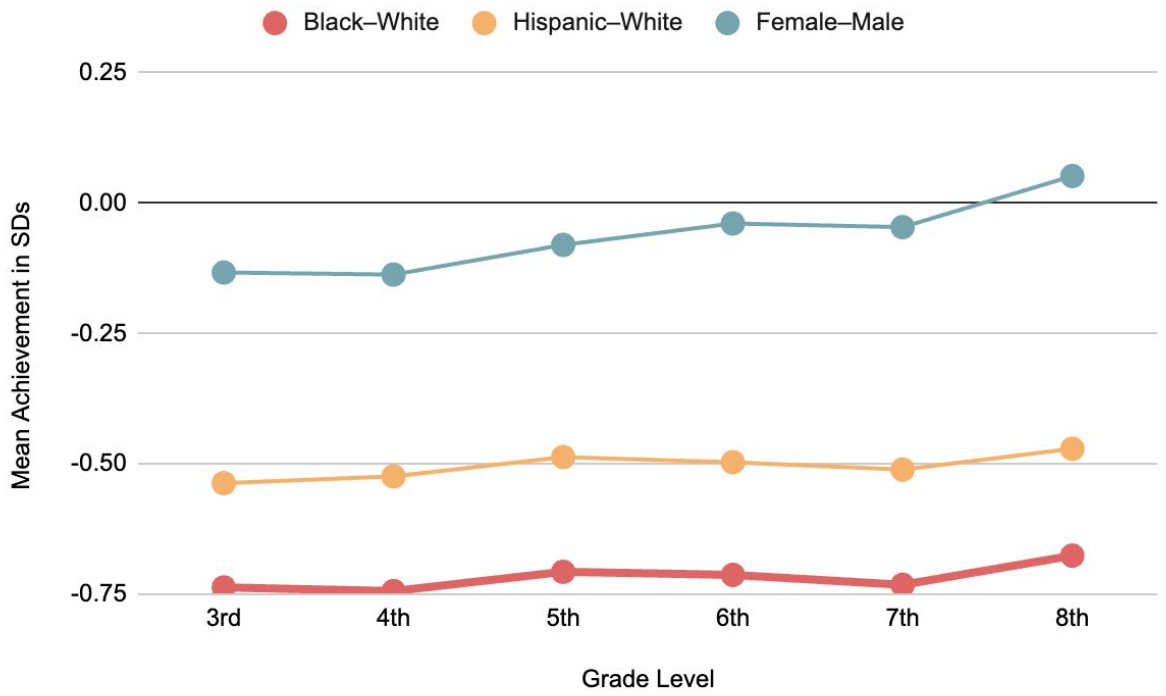


- Female–Male achievement gaps are smaller compared to the racial gaps but are still significant
- Black and Hispanic students are 2 and 1.2 years behind the national average respectively.

Student Group	Mean in SDs	Years to Avg	Current Status
All students	**-0.120**	-0.36	Slightly Below
White	**0.084**	0.25	Slightly Above
Black	**-0.697**	-2.09	Largest deficit
Hispanic	**-0.412**	-1.24	Moderate deficit
Asian	**0.711**	2.13	High performing
Female	**-0.133**	-0.39	Small deficit
Male	**-0.039**	-0.12	Close to average

A standard deviation (SD) of zero represents the national average standardized by SEDA based on the years 2009–2024 (Reardon et. al., 2036). 1 SD 3 years of learning (Hanushek, 2019)

Grade Level Achievement Gaps Post-COVID



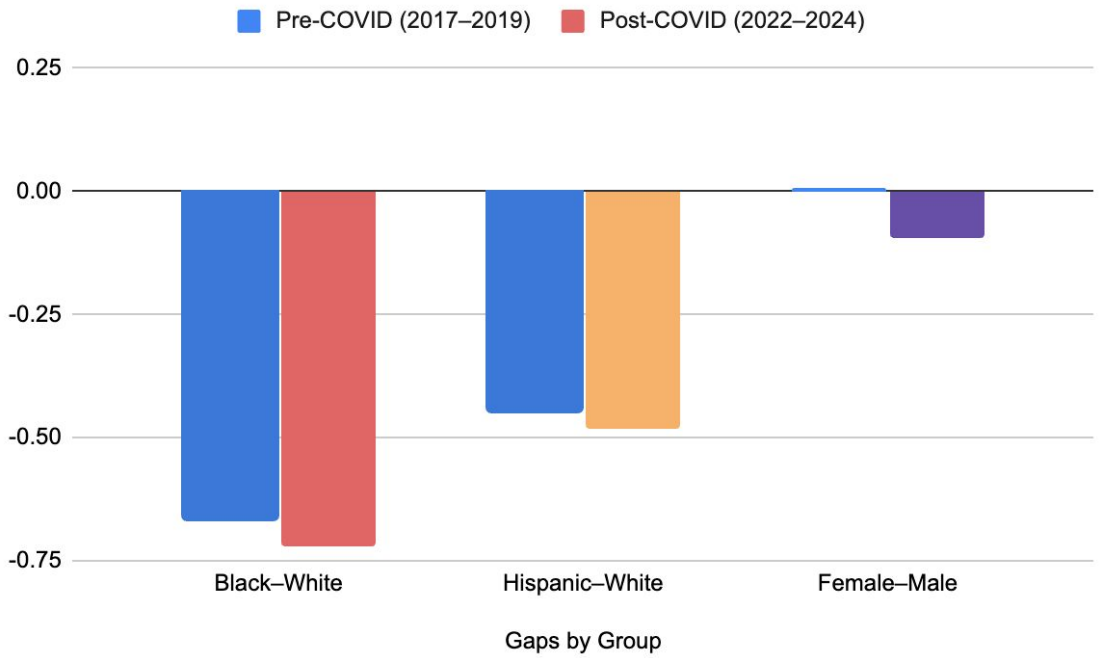
Female–Male:
The gap narrows by 8th grade. About a month or so head at +0.16 standard deviations.

Hispanic–White:
Around –0.50 SDs behind from 3rd–8th grade. Consistently about 1.5 years behind.

Black–White
Around –0.70 SDs from 3rd to 7th grade. The gap slightly closes by 8th grade. Consistently about 2 years behind.

Negative values indicate gaps between student group and the national average.

Post-COVID Math Achievement Gap Changes by Student Groups



Gaps have widened post-COVID:

Black-White:
The gap post-COVID has widened by 0.055 SDs or 31 school days lost.

Hispanic-White:
The gap post-COVID has widened by 0.036 SDs or 20 school days lost.

Female-Male:
The gap post-COVID has widened by 0.102 SDs or 56 school days lost. This is a reversal of pre-COVID (Kuhfeld, 2025).

What do these post-COVID gaps mean?

Black–White Gap	Hispanic–White Gap	Female–Male Gap
–0.70 SDs	**–0.48 SDs**	**–0.10 SDs**
= about 2 years behind White peers	= about 1.25 years of learning behind White peers	= about 18 days behind White male peers based on a 180 day school year. This is a reversal from pre-COVID

1. **Black and Hispanic students have different needs than Female students as whole. Targeted interventions are necessary.**
2. **Girls' participation in high level math courses has decreased post-COVID so a focus on encouraging girls to get into high level math classes could help improve this issue.**
3. **Racial gaps are a systemic emergency as well as ongoing recovery.**

The Role of Socioeconomic Status In Achievement Gaps

Research Question 2: To what extent are post-COVID math achievement gaps by race and gender affected by socioeconomic status, after controlling for grade level and year?

SES & Achievement Gaps

Gap	SES Coefficient	Interpretation	School Days	p-value	N
Black–White	-0.068	Gap widens by 0.068 SD per SES unit	37 days	< .001	26,207
Hispanic–White	-0.081	Gap widens by 0.081 SD per SES unit	44 days	< .001	43,666
Female–Male	-0.028	Even in higher-SES districts, girls fall behind	15 days	< .001	93,471

Key Findings:

- Male students disproportionately benefit from resources in higher SES districts compared to female students.
- Black and Hispanic students with similar socioeconomic status do not benefit or appear to have equal access to the same advantages as their White peers.

R^2 range: 4–8% across all models.

SES Changes from Pre- to Post-COVID

Gap	SES Coefficient	p-value	Significant?	N
Black–White	-0.009	0.01	***Yes***	9,458
Hispanic–White	0.005	0.103	No	15,227
Female–Male	0.008	< .001	***Yes***	34,806

Black–White Gap

Districts with a higher SES saw a wider gap post-COVID than districts with a lower SES. White students recovered much faster likely due to access to resources & the type of resources available.

Hispanic–White Gap

The gap widened post-COVID consistently across all SES levels for Hispanic students. The gaps were not exacerbated by wealth factors.

Female–Male Gap

Girls fell behind boys post-COVID relatively consistently across districts of all SES levels, however lower SES districts saw a slightly wider gap than higher SES districts on female students.

R^2 range: 4–8% across all models.

SES Changes from Pre- to Post-COVID

1

Racial Gaps:
Black students scored about 0.70 below White peers post-COVID putting them about 2 years behind. Hispanic students are about 1.25 years behind their White peers. These gaps have widened post-COVID across all grade levels.

2

Gender Gaps:
Gender Gaps reversed post-COVID with female students who were equal to male students pre-COVID, now behind by 0.095 standard deviations, which is about 56 school days. Lower SES districts were affected more than higher

3

Higher SES is inconsistent across race and gender:
Districts with a higher SES show a wider achievement gap. Black and Hispanic students benefit the least while White male students benefit the most. Placing students in higher SES districts is not a student outcome improvement solution.

Recommendations for District Leaders

District Leaders

- Treat racial achievement gaps as a systemic emergency & provide ongoing targeted interventions.
- Audit high SES districts to determine what resources are being utilized and incorporate resources that will close gaps for focus demographics.
- High dosage tutoring and summer school could improve outcomes if ongoing and consistent.

SES only explains a small fraction of what drives gaps so further predictive research is needed to be able to reliably forecast & begin prevention. .

Recommendations for State Education Agencies & Policymakers

State Education Agencies

- Fund recovery programs similar to ESSER to target both low & high SES districts and incorporate resources for students to close those gaps (high dosage tutoring, summer school, after school tutoring, hiring part time tutors, etc).
- Implement subgroup reporting to determine whether targeted interventions are working. This should be ongoing.
- Fund programs to encourage girls to enter higher level math courses.
- In high SES districts, provide remediation alongside advanced classes to target extension and recovery.

Policymakers

- Avoid programs that attempt to address every student, instead target subgroups.
- Invest in prevention from elementary to circumvent these systemic gap widenings including high quality preschool and math interventions by 3rd grade.
- Continue to research instructional strategies to prevent gap forming.

SES only explains a small fraction of what drives gaps so further predictive research is needed to be able to reliably forecast & begin prevention. .

Thank you!

Sources:

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